OPH Cosmology as a Finite-Source Prediction Program

Source, Transfer, and Likelihood Boundaries

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Abstract

Observer-patch holography supplies finite observer records and conditional local gravity and matter structures. It does not supply a physical primordial source, dark abundance, relativistic dark stress, or Boltzmann initial state. This paper states the interfaces required to turn finite OPH data into CMB and large-scale-structure observables. Existing spectra and sky comparisons are diagnostics. Cosmology is excluded from the OPH falsification program because the source, transfer, and likelihood derivations are immature.

1 Claim Boundary

A physical cosmological prediction requires one construction that maps finite OPH records to a covariant source, propagates that source through a declared background, and evaluates the resulting observables with a complete likelihood. No such construction is supplied. Simulator outputs may test numerical interfaces and expose incompatible assumptions. They carry no cosmological verdict.

The relevant source papers are `cosmology/oph_dark_matter_paper.tex`, `cosmology/oph_cosmological_vacuum_and_structure_formation.tex`, and `paper/recovering_relativity_and_standard_model_structure_from_observer_overlap_consistency_compact.tex`.

2 Finite-Source Interface

Let X_r denote a finite observer-screen state at refinement level r. A cosmological source map must emit

$$\mathsf{S}_r(X_r) = \left(g_{\mu\nu}, T_{\mu\nu}, \mathcal{I}, \mathcal{B}\right)_r, \tag{1}$$

where $T_{\mu\nu}$ is conserved, $\mathcal{I}$ contains gauge-invariant initial data, and $\mathcal{B}$ states the background and boundary conditions. The map must be defined without using the observed sky quantities that its outputs are compared with.

The screen-to-volume step requires a radial lift. Angular screen covariance alone does not determine a three-dimensional spectrum. The physical source map must therefore specify the radial measure, null-space treatment, clock, and normalization. These objects are work in progress.

3 Dark-Sector Interface

The dark-matter paper proposes a repair-charge rotor with occupation n and phase θ. Conditional on that action, a source-free dilute branch has

$$n \propto a^{-3}, \qquad \rho_R \propto a^{-3}, \qquad w_R \simeq 0. \tag{2}$$

Its cubic condensed branch gives the deep radial-acceleration law and the baryonic Tully–Fisher scaling under static spherical premises.

Equation (2) does not determine the abundance, initial perturbations, relativistic stress, gravitational slip, sound speed, nonlinear evolution, or coupling to the visible sector. A cosmological use requires a covariant completion that emits these quantities from the same action and obeys the full energy-momentum ledger. That completion is work in progress.

4 Primordial Interface

A physical primordial branch must supply all of the following:

1. a finite source ensemble and a clock;
2. a conserved covariant stress tensor;
3. gauge-invariant scalar, vector, and tensor initial data;
4. a screen-to-volume lift with a controlled null space;
5. phase, amplitude, and isocurvature conditions;
6. refinement stability and an error model.

Candidate formulas involving the screen scalar P, including spectral-tilt expressions, are diagnostic until this interface is constructed. Numerical agreement with a measured summary does not supply the missing source map.

5 Transfer Interface

Given a completed source map, the transfer calculation must declare the background species, equations of state, collision terms, recombination model, initial modes, gauge convention, numerical tolerances, and observable projection. Standard Einstein–Boltzmann software can test this interface with imported inputs. Such a run tests the solver path rather than an OPH source theorem.

The minimum transfer outputs are

$$\left(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, P_m(k,z), H(z), D_A(z), f\sigma_8(z)\right). \tag{3}$$

All outputs must arise from one compatible background and perturbation system.

6 Likelihood Interface

A data comparison must state the datasets, masks, covariances, nuisance parameters, priors, and combination rule. Overlapping samples and shared calibrations must be represented in the covariance model. Visual overlays, diagonal residual sums, and comparisons made after selecting a formula are diagnostics.

The current cosmology materials contain diagnostic CMB, galaxy, and background comparisons. They do not define cosmological falsifiers or forward targets.

7 Current Status

Object	Status
Finite observer-screen carriers	Conditional finite construction
Screen scalar covariance	Conditional source-level construction
Physical radial lift	Work in progress
Primordial covariant source	Work in progress
Repair-charge cosmological abundance	Work in progress
Relativistic repair stress and slip	Work in progress
Boltzmann initial-state bridge	Work in progress
Joint CMB, BAO, lensing, growth, and cluster likelihood	Work in progress
Cosmological falsification target	Excluded from the program

8 Conclusion

Finite OPH screen data become cosmology only through a covariant source map, a three-dimensional lift, a conserved stress tensor, compatible transfer equations, and a complete likelihood. These constructions are work in progress. Existing cosmological calculations are diagnostics and carry no falsification verdict.